

HAMBURG FUHLSBUTTEL, Germany

WMO: 101470

Lat: 53.6331N

Lon: 9.9881E

Elev: 15

StdP: 101.14

Time Zone: 1.00 (EUC)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.2	-6.1	-12.1	1.3	-6.8	-9.2	1.7	-4.8	12.2	7.2	11.1	7.2	3.5	100	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	29.1	19.2	27.1	19.0	25.1	17.9	20.7	26.6	19.7	25.2	18.8	23.5	4.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.9	13.7	22.6	17.9	12.9	21.7	17.0	12.2	20.8	59.4	26.8	56.3	25.2	53.4	23.6	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.0	8.9	8.0	-11.1	33.1	3.3	2.4	-13.5	34.9	-15.5	36.3	-17.3	37.6	-19.7	39.4
			-11.5	22.8	3.4	0.7	-13.9	23.3	-15.9	23.8	-17.8	24.2	-20.2	24.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	1.6	2.0	4.9	9.6	13.0	16.2	18.7	17.8	15.0	10.7	6.2	3.7
	DBStd	6.96	4.39	3.76	4.22	3.75	3.83	3.22	3.31	2.79	2.83	3.30	3.70	4.17
	HDD10.0	1062	259	224	162	52	13	0	0	0	1	31	121	198
	HDD18.3	3149	518	458	417	263	172	80	36	44	107	237	363	455
	CDD10.0	1056	0	0	4	39	105	186	269	241	150	53	8	1
	CDD18.3	101	0	0	0	0	6	16	47	26	6	0	0	0
	CDH23.3	874	0	0	0	6	57	154	421	195	40	2	0	0
	CDH26.7	234	0	0	0	0	10	33	135	52	4	0	0	0
Wind	WSAvg	4.0	4.7	4.4	4.4	4.1	4.0	3.8	3.6	3.5	3.6	3.9	4.1	4.6
Precipitation	PrecAvg	774	64	46	56	46	58	75	81	76	67	62	68	74
	PrecMax	1073	134	133	150	114	160	210	174	236	180	183	144	175
	PrecMin	490	1	6	6	1	5	11	5	8	11	16	8	12
	PrecStd	128	34	27	33	25	30	40	42	38	35	34	32	42
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.8	12.9	18.0	24.0	28.0	29.8	33.1	31.1	27.2	22.0	16.0	13.0
		MCWB	10.2	7.4	11.0	14.3	18.4	19.5	20.9	20.3	19.3	16.5	13.5	11.0
	2%	DB	10.2	10.0	15.0	21.1	24.8	27.2	30.1	27.8	24.1	18.9	14.0	11.8
		MCWB	9.0	8.1	10.1	13.1	17.0	19.1	19.7	19.3	17.8	15.5	12.4	10.2
	5%	DB	8.9	8.2	12.2	19.0	22.2	25.0	27.9	25.1	22.0	16.9	12.8	10.2
		MCWB	8.1	6.7	8.9	12.4	15.0	17.6	19.1	18.1	16.8	14.3	11.2	9.0
	10%	DB	7.8	7.0	10.8	16.8	20.1	22.8	25.8	23.2	19.9	15.2	11.1	9.0
		MCWB	7.0	5.8	8.1	11.5	14.5	16.6	18.5	17.3	15.8	13.2	9.6	8.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.8	9.4	12.6	15.3	19.7	21.3	22.2	21.5	19.7	17.1	14.4	11.7
		MCDB	11.7	10.1	16.0	21.7	26.4	26.8	28.9	27.3	24.7	20.7	15.7	12.9
	2%	WB	9.3	8.0	10.8	13.9	17.8	19.8	20.9	20.2	18.5	15.6	12.5	10.5
		MCDB	10.0	9.5	13.3	19.7	22.7	25.7	27.3	25.8	23.1	18.2	13.9	11.5
	5%	WB	7.9	6.8	9.6	12.9	16.3	18.5	20.0	19.2	17.4	14.6	11.3	9.3
		MCDB	8.7	8.1	12.1	17.6	21.1	23.1	25.6	23.8	21.0	16.6	12.7	10.2
	10%	WB	6.7	5.7	8.2	11.7	14.9	17.4	19.0	18.2	16.4	13.5	10.1	7.8
		MCDB	7.3	6.8	10.3	16.2	18.9	21.6	24.1	22.1	19.5	15.3	11.0	8.6
Mean Daily Temperature Range	5% DB	MDBR	4.2	4.9	7.1	9.3	9.4	9.5	9.3	8.8	8.0	6.5	4.7	4.1
		MCDBR	4.6	6.5	10.5	13.4	13.1	13.4	13.5	12.1	11.4	8.5	5.4	4.9
		MCWBR	4.5	5.0	6.5	7.0	6.7	6.7	5.3	5.5	6.1	5.5	4.4	4.2
	5% WB	MCDBR	4.4	6.1	9.2	11.9	11.7	11.6	11.4	10.3	9.5	7.7	5.3	4.8
		MCWBR	4.4	5.0	6.2	6.9	6.5	6.3	5.2	5.5	5.6	5.3	4.4	4.4
Clear-Sky Solar Irradiance	taub	0.318	0.337	0.367	0.419	0.412	0.409	0.428	0.421	0.392	0.356	0.334	0.310	
	taud	2.344	2.348	2.290	2.180	2.243	2.277	2.240	2.272	2.336	2.427	2.390	2.339	
	Ebn at Noon	644	756	807	806	836	841	815	797	773	727	621	576	
	Edh at Noon	58	79	102	130	128	126	128	118	98	73	56	49	
All-Sky Solar Radiation	RadAvg	0.56	1.24	2.43	4.02	5.05	5.30	5.02	4.15	3.02	1.66	0.70	0.42	
	RadStd	0.07	0.17	0.24	0.51	0.59	0.41	0.68	0.31	0.34	0.19	0.07	0.03	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-143	N/A	N/A

Nomenclature: See separate page